🌐 1️⃣ What is Azure DNS?

Azure DNS is Microsoft's DNS hosting service.

There are **two major types**:

Type	Used For
Public DNS	Internet-facing domains
Private DNS	Internal name resolution inside VNet

🌍 2️⃣ Azure Public DNS

Used to host:

```
example.com
```

Azure becomes authoritative DNS provider.

You create:

- A record
- CNAME
- MX
- TXT
- NS
- AAAA

🔥 Public DNS Lab

Step 1 – Create DNS Zone

Portal → Search → DNS Zones → Create

- Name: `yourdomain.com`
- Resource group: same region

Step 2 – Add A Record

DNS Zone → Record Sets → Add

Setting	Value
Name	www
Type	A
IP Address	Your VM Public IP

Save.

Step 3 – Update Domain Registrar

Update NS records at:

- GoDaddy
- Namecheap
- Any domain provider

Point to Azure name servers.

Step 4 – Test

```nginx
nslookup www.yourdomain.com
```

Should resolve to VM public IP.

🔐 3️⃣ Azure Private DNS (Very Important for AZ-104)

Used for:

- Internal name resolution
- VNet resources
- Private Endpoints
- Hybrid network

Example:

```pgsql
web.internal.local
```

Not visible on Internet.

🔥 Private DNS Lab

🎯 Scenario

Two VMs in same VNet.

Access by hostname instead of IP.

Step 1 – Create Private DNS Zone

Portal → Private DNS Zones → Create

Name:

```pgsql
internal.local
```

Step 2 – Link VNet

Private DNS Zone → Virtual Network Links → Add

Select:

Your VNet

Enable:

✔️ Auto-registration (optional)

Step 3 – Add A Record

Add record:

Name	web
IP	10.0.1.4

Now DNS record:

```pgsql
web.internal.local
```

Step 4 – Test from VM

Inside another VM:

```pgsql
ping web.internal.local
```

Should resolve private IP.

📊 Public vs Private DNS

Public DNS	Private DNS
Internet-facing	Internal only
Requires domain ownership	No domain ownership required
Hosted in Azure	Linked to VNet
Global resolution	VNet scope

🧠 4️⃣ Azure Default DNS

By default:

Every VNet uses:

```
168.63.129.16
```

Azure-provided DNS server.

It resolves:

- VM hostnames
- Private DNS Zones
- Azure services

🏗️ 5️⃣ Custom DNS

If using:

- Active Directory Domain Services
- On-prem DNS
- Hybrid network

You can configure:

VNet → DNS Servers → Custom

Example:

```scss
10.0.0.4 (Domain Controller)
```

All VMs will use that DNS.

🔥 6️⃣ Important AZ-104 Concepts

- ✅ Private DNS works only inside linked VNet
- ✅ Can link multiple VNets
- ✅ Private DNS supports auto-registration
- ✅ Public DNS requires domain ownership
- ✅ DNS zone is global resource
- ✅ DNS does not filter traffic (only resolves names)

🧪 7️⃣ Troubleshooting DNS

Inside VM:

```bash
ipconfig /all
nslookup hostname
```

Check:

- Which DNS server is used
- Does record exist
- Is VNet linked